

TRADEPASS ACADEMY

309A COMPANION SERIES · VOLUME 2

Last-Minute Exam Review

The final-week condensation of the Construction Electrician 309A exam: 100 facts, memory tricks, traps and a revision plan for the last 48 hours.

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- The top 100 exam facts, grouped by topic
 - Memory tricks that survive exam-room stress
 - The mistakes that cost strong candidates marks
 - Frequently tested concepts, condensed
 - Question traps and how to disarm them
 - A final 48-hour revision checklist

Companion resource to: Red Seal Construction Electrician 309A Exam Prep, 2026–2027 Edition

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How to use this review

This volume is engineered for the final seven days before your 309A sitting. It assumes the study work is done: you have been through your TradePass Academy guide, written the practice tests, and know your weak areas. What remains is consolidation, and that is what this document delivers.

- **Days 7–4 before the exam:** read one fact group per sitting (Section 1) and one frequently tested concept (Section 4). Mark anything that surprises you; surprises are study targets.
- **Days 3–2:** work the memory tricks (Section 2), common mistakes (Section 3) and exam traps (Section 5) in full. Do the quick calculations (Section 6) until they are automatic.
- **Final 48 hours:** follow the checklist in Section 7 exactly. No new material after the final evening.

KEY POINT

Code-referenced values in this review reflect long-standing CEC requirements, but the code is amended on a regular cycle and provinces adopt editions on their own schedules. Where a question depends on an exact code number, always answer from the edition in force for your examination.

1. The top 100 exam facts

1.1 Safety and fundamentals (facts 1–10)

#	Fact to know cold
1	Verify de-energization with the test-verify-test method: prove your meter on a known live source, test the circuit, then prove the meter again.
2	Current, not voltage, causes injury: currents in the range of tens of milliamperes across the chest can be fatal.
3	A Class A GFCI trips on a ground-fault imbalance of approximately 4 to 6 milliamperes.
4	The Canadian Electrical Code governs installations; CSA Z462 governs safe work practices around energized equipment.
5	Each province or territory adopts the CEC with its own amendments; the local inspection authority has the final word.
6	Arc-flash PPE is selected from an incident-energy analysis or the applicable CSA Z462 tables, never by guesswork.
7	Standard Canadian nominal voltages: 120/240 V single-phase; 120/208 V and 347/600 V three-phase; 60 Hz throughout.
8	Equipment must bear the mark of an accredited certification body (such as CSA or cUL) or be specially approved for the installation.
9	Portable ladders used near exposed energized parts must be of nonconductive material.
10	The first action in every incident scenario is to eliminate the hazard to yourself before assisting others: de-energize, then rescue.

1.2 DC circuit theory (facts 11–20)

#	Fact to know cold
11	Ohm's law: $E = I \times R$; know all three arrangements without hesitation.
12	In a series circuit the current is the same everywhere and the source voltage divides across the loads.
13	In a parallel circuit the voltage is the same across every branch and the current divides inversely with resistance.
14	Adding any resistance in parallel always lowers the total resistance of the circuit.
15	Heating in a conductor follows $P = I^2R$: doubling the current quadruples the heat.
16	Kirchhoff's voltage law: around any closed loop, the sum of the voltage drops equals the source voltage.
17	Kirchhoff's current law: the current entering a junction equals the current leaving it.
18	A voltmeter connects in parallel and has very high internal resistance; an ammeter connects in series and has very low resistance.
19	Across an open in a series circuit, the full source voltage appears; the rest of the circuit reads zero volts drop.
20	A short circuit bypasses load resistance, so current is limited only by the remaining impedance of the system.

1.3 AC theory (facts 21–30)

#	Fact to know cold
21	Meters read RMS values; RMS is the effective (heating-equivalent) value of an AC waveform.
22	Peak voltage = $\text{RMS} \times 1.414$; a 120 V RMS circuit peaks near 170 V.
23	In an inductor, current lags voltage; in a capacitor, current leads voltage (ELI the ICE man).
24	Inductive reactance $X_L = 2\pi fL$ rises with frequency; capacitive reactance $X_C = 1/(2\pi fC)$ falls with frequency.
25	Power factor = true power $\div$ apparent power = $\cos \theta$, and can never exceed 1.
26	Capacitors are added to a system to correct a lagging (inductive) power factor.
27	At series resonance $X_L = X_C$, impedance is at its minimum and equals R, and current is at its maximum.
28	Only resistance consumes true power; ideal inductors and capacitors exchange energy without consuming it.
29	The $\sqrt{3}$ factor (1.732) appears in every three-phase line calculation; forgetting it is the most common calculation error on the exam.
30	Reversing any two line conductors reverses the rotation of a three-phase motor.

1.4 Conductors and cables (facts 31–40)

#	Fact to know cold
31	Conductor ampacity depends on insulation temperature rating, ambient temperature, and the number of current-carrying conductors grouped together.
32	The termination temperature rating (commonly 75 °C) can limit the usable ampacity below the insulation rating.
33	When more than three current-carrying conductors share a raceway or cable, ampacity must be derated per the code tables.
34	For equal ampacity, aluminum requires a larger size than copper and must use terminations approved for aluminum.
35	Small-conductor overcurrent limits to memorize: #14 Cu $\rightarrow$ 15 A, #12 Cu $\rightarrow$ 20 A, #10 Cu $\rightarrow$ 30 A.
36	Green (or green-yellow) identifies bonding conductors only; white or grey identifies the neutral (identified conductor).
37	General raceway fill for three or more conductors is 40 percent of cross-sectional area.
38	Continuous loads are limited to 80 percent of the overcurrent device rating unless the assembly is rated for 100 percent.
39	On 120/208 V systems feeding electronic (nonlinear) loads, the neutral can carry harmonic current approaching or exceeding phase current.
40	A conductor's resistance rises with temperature; ratings and calculations state their reference temperature for this reason.

1.5 Wiring methods (facts 41–50)

#	Fact to know cold
41	Nonmetallic sheathed cable (NMD90) is for dry locations and must be protected where subject to mechanical damage.
42	Cables run through studs need approximately 32 mm of edge clearance or a protective steel plate.
43	Support nonmetallic sheathed cable at regular intervals (about every 1.5 m) and within 300 mm of every box.
44	Every splice must remain accessible; junction boxes may never be concealed behind finishes.
45	Box fill is counted in conductor equivalents, and a device (receptacle or switch) counts as two conductors.
46	EMT joins with approved fittings, not threads, and in many installations serves as the bonding path when properly assembled.
47	Long PVC conduit runs exposed to temperature change require expansion fittings.
48	Wet locations require fittings and covers rated for the exposure; “rain-tight” and “wet location” markings matter.
49	Direct-buried conductors and raceways have minimum cover depths that increase where vehicles pass.
50	Raceways are supported at code-specified intervals and near every box, fitting and termination.

1.6 Grounding and bonding (facts 51–60)

#	Fact to know cold
51	Grounding connects the system to the earth; bonding connects metal parts together so fault current can flow and trip the protection. They are different questions on the exam.
52	The neutral is grounded at the service (system grounding point) and kept separate from bonding conductors everywhere downstream.
53	In a sub-panel, the neutral bar is isolated from the enclosure and from the bond bar.
54	The fault-return path must be low impedance so the overcurrent device opens quickly; a poor bond means a slow or absent trip.
55	Bonding conductor size is selected from the code table based on the rating of the overcurrent device ahead of it.
56	Where ground rods are used, they are typically installed as a pair spaced at least 3 m apart.
57	A GFCI protects by detecting current imbalance between live and neutral; it functions even with no bonding conductor present.
58	Concentric or eccentric knockouts at services require bonding jumpers or approved fittings to guarantee the fault path.
59	An isolated-ground receptacle still requires an insulated bonding conductor run back with the circuit; it is never truly “unbonded.”
60	The green bonding screw of a device and the box must complete a continuous path to the panel bond bar.

1.7 Distribution and protection (facts 61–70)

#	Fact to know cold
61	The interrupting rating of a breaker or fuse must equal or exceed the available fault current at its point of installation.
62	Time-delay fuses ride through motor inrush; HRC fuses provide high interrupting capacity common in Canadian gear.
63	A working space of about 1 m must be kept clear in front of electrical equipment requiring operation or maintenance.
64	The main bonding jumper at the service is the single point where neutral and bonding systems join.
65	Demand factors allow calculated service and feeder loads to be smaller than the connected total, reflecting real usage.
66	Overcurrent protection protects conductors; it is sized to the conductor unless a specific rule permits otherwise (motors, welders, taps).
67	Standard overcurrent device ratings follow a fixed series (15, 20, 30, 40, 50 A ...); many questions turn on choosing the next standard size.
68	Selective coordination means the device nearest the fault opens first, keeping the rest of the system alive.
69	A kilowatt-hour meter records energy; a demand meter records the peak rate of use for billing.
70	Fuses are single-use and single-pole by nature; multi-pole breakers open all poles together, which matters for multi-wire circuits.

1.8 Motors and controls (facts 71–80)

#	Fact to know cold
71	The branch protection (fuse/breaker) covers short circuits and ground faults; the overload relay covers running overcurrent. Two different protections, two different sizings.
72	Overload protection is sized from the motor nameplate full-load current, adjusted for service factor as the code permits.
73	Motor branch-circuit conductors are sized at a minimum of 125 percent of the motor full-load current.
74	For installation sizing, full-load current comes from the code tables by horsepower and voltage, not from calculation.
75	Synchronous speed $n_s = 120f \div P$; at 60 Hz: 2 poles $\rightarrow$ 3600, 4 $\rightarrow$ 1800, 6 $\rightarrow$ 1200, 8 $\rightarrow$ 900 r/min.
76	Induction motors run at 2–5 percent slip below synchronous speed at full load.
77	Across-the-line starting draws roughly six times full-load current; reduced-voltage and soft starters cut the inrush.
78	Three-wire control (start-stop with seal-in contact) gives low-voltage protection: the motor stays off after an outage until restarted.
79	Stop devices and safety interlocks wire in series (normally closed); multiple start devices wire in parallel (normally open).
80	Every motor requires a disconnecting means; it must be within sight of the motor or capable of being locked in the open position.

1.9 Transformers (facts 81–90)

#	Fact to know cold
81	Voltage follows the turns ratio; current follows the inverse of the turns ratio; power in equals power out (less losses).
82	Transformers are rated in kVA because winding heating depends on current regardless of load power factor.
83	A delta-wye transformer stepping 600 V down produces the familiar 120/208 V system with a neutral.
84	Primary and secondary overcurrent protection percentages come from the code tables; know that both windings may need protection.
85	The full-load current of a three-phase transformer: $I = \text{kVA} \times 1000 \div (\sqrt{3} \times \text{EL})$. Memorize 45 kVA at 208 V $\rightarrow$ 125 A.
86	Transformer polarity (additive or subtractive) must match before paralleling units.
87	Never open the secondary of a current transformer while the primary is energized; dangerous voltage develops instantly.
88	Dry-type transformers need ventilation clearance; blocked airflow is a common overheating cause.
89	Percent impedance sets the available secondary fault current: lower %Z means higher fault current.
90	An autotransformer shares one winding between primary and secondary: smaller and cheaper, but it provides no isolation.

1.10 Lighting, systems and special installations (facts 91–100)

#	Fact to know cold
91	LED retrofit lamps must be compatible with the existing ballast or driver arrangement stated by the manufacturer; “ballast-bypass” changes the luminaire’s certification.
92	Self-contained emergency lighting (unit equipment) carries its own battery and must be connected ahead of local switching.
93	Fire-alarm wiring classes differ in supervision: Class A returns to the panel so a single open does not disable devices; Class B does not.
94	Class 2 low-voltage circuits are power-limited and must be separated from line-voltage conductors unless barriers or permitted exceptions apply.
95	Dwelling receptacles are generally required to be tamper-resistant.
96	Outdoor receptacles require GFCI protection and covers suitable for the exposure, including in-use covers where equipment stays plugged in.
97	Kitchen counter receptacles carry dedicated branch-circuit rules that have changed across code editions; answer from the edition in force.
98	Smoke alarms in dwellings must be interconnected and provided with battery backup under the building code.
99	Communications and data cabling is separated from power wiring to limit induced noise and for safety.
100	Photovoltaic conductors are treated as energized whenever the array is illuminated; rapid-shutdown and disconnect labelling requirements apply.

2. Memory tricks that survive exam pressure

A mnemonic earns its place only if it works when you are nervous. These are the ones electricians actually use.

Trick	What it locks in
ELI the ICE man	In an inductor (L), E leads I. In a capacitor (C), I leads E.
"A horse walks in with 746 watts."	1 hp = 746 W.
"36-18-12-9"	Synchronous speeds in hundreds for 2, 4, 6, 8 poles at 60 Hz.
The letter Y has a centre point	A wye connection has a neutral point; a delta does not.
"Volts divide in series; amps divide in parallel."	Which quantity splits in which circuit.
"Breakers loaf at eighty."	Continuous loads: 80 percent of device rating.
"Motors get a quarter more."	Motor conductors: minimum 125 percent of full-load current.
"Stops in series, starts in parallel."	Control wiring of NC stop and NO start devices.
"Grounding goes to ground; bonding bonds the boxes."	The grounding vs bonding distinction.
"Four to six keeps hearts in ticks."	Class A GFCI trip range: 4-6 mA.
"Seventeen-thirty-two, three-phase for you."	$\sqrt{3} = 1.732$ in every three-phase line formula.
"Swap two, reverse the rev."	Reversing any two lines reverses a three-phase motor.

3. The mistakes that cost marks

These fifteen errors account for a large share of lost marks among candidates who otherwise know the material. Read each one and recall the last time you made it in practice.

#	Mistake	The fix
1	Dropping the $\sqrt{3}$ in three-phase calculations	Write the formula before substituting numbers; the formula carries the $\sqrt{3}$ for you.
2	Confusing line and phase quantities in wye and delta	Sketch the connection. Wye: $E_L = \sqrt{3} E_\phi$. Delta: $I_L = \sqrt{3} I_\phi$.
3	Sizing motor conductors from nameplate current	Installation sizing uses the code table value by hp and voltage.
4	Mixing up overload and short-circuit protection	Overloads protect the motor; the branch device protects against faults.
5	Dividing by power factor when multiplication was needed (or vice versa)	From power to current you divide by PF; from $V \times I$ to watts you multiply.
6	Using round-trip length as one-way length in voltage drop	The formulas already include the factor 2 (or $\sqrt{3}$); L is one-way.
7	Applying slip to the wrong base	Slip is a percentage of synchronous speed, not of shaft speed.
8	Using the two-resistor product/sum shortcut on three resistors	Product-over-sum works for exactly two branches; use reciprocals otherwise.
9	Forgetting ambient and grouping derating	Any question that mentions temperature or conductor count is a derating question.
10	Treating kVA as kW	$kW = kVA \times PF$. Watch which one the question gives and which it asks for.
11	Mixing RMS and peak values	Unless a question says peak, every AC value is RMS.
12	Assuming the neutral carries no current on 3 ϕ electronic loads	Harmonics can load the neutral heavily; questions test exactly this.
13	Bonding conductor sized from the wrong basis	Bonding follows the overcurrent device rating, from the bonding table.
14	Ignoring the 80 percent continuous limit	The word "continuous" in a question is the trigger to apply it.
15	Reporting an intermediate result as the final answer	Re-read what the question asked before selecting; distractors are built from your intermediate numbers.

4. Frequently tested concepts, condensed

4.1 Grounding versus bonding

The single most reliable topic on the exam. Grounding ties the electrical system to earth and stabilizes voltage; bonding ties all non-current-carrying metal together so that a fault becomes a large, breaker-tripping current instead of a lethal touch voltage. The neutral-to-bond connection happens once, at the service; every panel downstream keeps them separated.

4.2 GFCI and arc-fault protection

A GFCI compares current out on the live conductor with current back on the neutral and opens on a small imbalance; it protects people and needs no bonding conductor to work. Arc-fault devices recognize the signature of arcing and protect against fire. Expect questions on where each is required and on what each does and does not detect.

4.3 The motor protection stack

Four layers, four sizings: the disconnect (within sight or lockable), the branch short-circuit device (sized generously to ride through inrush), the conductors (minimum 125 percent of table FLC), and the overload relay (sized from nameplate). Most motor questions are simply asking which layer is being sized and from which number.

4.4 Three-phase systems

Know the two standard secondaries cold: 120/208 V wye and 347/600 V wye, plus delta configurations without a neutral. Every calculation runs through $\sqrt{3}$, and every connection question runs through the wye/delta relationships. Rotation reverses when any two lines are exchanged.

4.5 Transformer behaviour

Turns ratio sets voltage, inverse ratio sets current, kVA passes through. Protection percentages, polarity for paralleling, the CT-secondary hazard, and impedance's effect on fault current are the four details examiners reach for beyond the ratios.

4.6 Voltage drop

Two methods, one pair of limits: 3 percent for the final circuit, 5 percent overall. Metric questions hand you resistivity and area; imperial questions hand you K and circular mils. The factor at the front is 2 for single-phase and $\sqrt{3}$ for three-phase, and L is always one-way.

4.7 Ampacity and derating

Start from the table value for the insulation, then apply ambient correction and grouping derating, then check the termination rating, then confirm against the overcurrent limit for small conductors. Questions usually hide exactly one of these steps.

4.8 Box fill and conduit fill

Box fill counts conductor equivalents, with devices counting as two and all bonds together as one; conduit fill for three or more conductors is 40 percent. Expect one straightforward counting question; the marks are lost through rushing, not difficulty.

4.9 Control circuits

Two-wire control restarts automatically after an outage (low-voltage release); three-wire control with a seal-in contact stays off (low-voltage protection). Safety devices and stops are normally closed in series; starts are normally open in parallel. Be able to trace a simple ladder diagram quickly.

4.10 Services and demand

Calculated load is not connected load: demand factors recognize diversity. Know the flow of a basic calculation, the purpose of the main bonding jumper, working clearances, and that the interrupting rating must meet available fault current.

5. Exam traps and how to disarm them

Trap	Disarm it
EXCEPT / NOT / LEAST in the stem	Underline the reversal word and restate the question in your own words before reading the options.
“Minimum” versus “maximum”	Code questions hinge on which limit is asked; circle the word.
Mixed units in the data	Convert everything to base units before calculating; mm vs m and kW vs W are the classic setups.
“Line” versus “phase” wording	Sketch a five-second wye or delta before answering.
“What should you do FIRST”	Safety and de-energization outrank diagnosis; sequence questions reward the safe order, not the fast one.
Distractors built from intermediate results	Finish the calculation, then re-read the question before choosing.
Copper values offered on an aluminum question	Check the material every time resistivity or K appears.
“Most nearly” answers	Carry an extra decimal through the work and round only at the end.
Unstated assumptions	Three-phase systems are balanced and voltages nominal unless the question says otherwise.
Edition-dependent numbers	Where options differ by code edition, answer from the edition your exam is based on.

6. Quick calculations to make automatic

Drill these until they need no paper. They convert calculation questions from two minutes to twenty seconds.

Shortcut	Value	Example
Amps at 120 V	$W \div 120$	1 500 W $\rightarrow$ 12.5 A
Amps at 240 V	$W \div 240$	4 800 W $\rightarrow$ 20 A
3 ϕ divisor at 208 V	$\sqrt{3} \times 208 \approx 360$	18 kW, PF 1 $\rightarrow$ 50 A
3 ϕ divisor at 600 V	$\sqrt{3} \times 600 \approx 1\,039$	15 kW, PF 1 $\rightarrow$ 14.4 A
1 kVA of load draws	8.3 A @120 • 4.2 A @240 • 2.8 A @208 3 ϕ • 0.96 A @600 3 ϕ	scale linearly
Synchronous speeds	3600 / 1800 / 1200 / 900	for 2 / 4 / 6 / 8 poles
Peak from RMS	$\times 1.414$	347 V $\rightarrow$ $\approx$ 490 V peak
hp to watts	$\times 746$	10 hp $\rightarrow$ 7 460 W
125% in your head	add a quarter	28 A $\rightarrow$ 35 A
80% in your head	subtract a fifth	40 A device $\rightarrow$ 32 A continuous

8. Definitions in one line

The exam tests vocabulary directly and hides it inside stems. One line each; if any line reads as new information, that term goes on your Day 28 repair list.

Term	One-line definition
Ampacity	The maximum current a conductor can carry continuously under its conditions of use.
Bonding	Connecting non-current-carrying metal parts together to form a low-impedance fault path.
Grounding	Connecting the electrical system or equipment to the general mass of the earth.
Identified conductor	The grounded circuit conductor (neutral), coloured white or grey.
Overcurrent	Any current above rating: overload, short circuit or ground fault.
Overload	An overcurrent confined to the normal circuit path, from excess load rather than a fault.
Short circuit	An abnormal, low-impedance connection between conductors of different potential.
Ground fault	An unintended connection between an ungrounded conductor and ground or bonded metal.
Interrupting rating	The highest fault current a protective device can safely open.
Continuous load	A load expected to persist at maximum current for a sustained period.
Demand factor	The ratio of calculated demand to total connected load.
Power factor	True power divided by apparent power; the cosine of the phase angle.
Impedance	Total AC opposition to current: resistance and reactance combined.
Reactance	Opposition to AC from inductance or capacitance, in ohms.
RMS value	The effective value of an AC quantity, equal in heating effect to the same DC value.

Term	One-line definition
Slip	The percentage by which rotor speed falls below synchronous speed.
Service factor	The permissible overload multiple a motor can sustain, marked on the nameplate.
Full-load current	The current a motor draws at rated load, voltage and frequency.
Turns ratio	Primary turns divided by secondary turns; sets the voltage ratio.
Percent impedance	A transformer's internal impedance expressed as a percentage; governs fault current.
Wye connection	Three windings joined at a common neutral point.
Delta connection	Three windings joined end-to-end in a closed triangle, no neutral.
Class 2 circuit	A power-limited low-voltage circuit with relaxed wiring rules and required separation.
Disconnecting means	A device by which circuit conductors can be disconnected from their supply.

9. Rapid-fire self-quiz

Cover the right column. Twenty answers in five minutes means this volume has done its job; any hesitation names your next review target.

#	Question	Answer
1	Trip range of a Class A GFCI?	4–6 mA
2	Watts in one horsepower?	746 W
3	Synchronous speed, 4 poles, 60 Hz?	1 800 r/min
4	Peak of 120 V RMS?	≈ 170 V
5	Continuous load limit on a 40 A breaker?	32 A
6	Motor conductor minimum, 28 A FLC?	35 A
7	Max overcurrent for #12 copper?	20 A
8	Line voltage of a wye with 120 V phases?	208 V
9	Reverse a 3φ motor how?	Swap any two line leads
10	Which leads in a capacitor: E or I?	Current leads
11	General raceway fill, 3+ conductors?	40 percent
12	Where do neutral and bond connect?	Once, at the service
13	kW from 50 kVA at 0.9 PF?	45 kW
14	Voltage-drop limits, branch / total?	3% / 5%
15	3φ divisor at 600 V?	≈ 1 039
16	Stops wire in ... ? Starts wire in ... ?	Series; parallel
17	Which control scheme auto-restarts?	Two-wire
18	CT secondary rule?	Never open under load
19	Full-load current, 45 kVA at 208 V 3φ?	125 A
20	A box splice may be concealed: true or false?	False; must stay accessible

10. The final 48 hours

10.1 Two days before

- ☐ Rewrite the formula index from memory; check it against Volume 1 and re-drill any gaps.
- ☐ Re-read your marked facts from Section 1 of this volume only; do not re-open full chapters.
- ☐ Work ten calculation questions from your weakest topic, timed at one minute each.
- ☐ Confirm exam location, start time, required identification and permitted materials with your authority.
- ☐ Prepare everything physical tonight: ID, confirmation, approved calculator, water, directions.

10.2 The day before

- ☐ Morning: one final timed set of 20 mixed questions. Stop regardless of score; the result is information, not a verdict.
- ☐ Afternoon: memory tricks (Section 2) and quick calculations (Section 6) once through, out loud.
- ☐ Evening: no study after dinner. Sleep is the highest-value preparation remaining.

10.3 Exam morning

- ☐ Eat, hydrate, arrive early enough to sit calmly before entry.
- ☐ On starting: skim the whole paper for thirty seconds so its shape holds no surprises.
- ☐ Budget roughly one minute per question on the first pass; flag and move rather than stall.
- ☐ Answer every question; an unanswered question is a guaranteed zero.
- ☐ Expect nerves in the first five questions and breathe through them; your practice tests were built to be at least this hard.

LAST WORD

You have prepared with the same standard the exam is built from. Trust the preparation, read every stem twice, and go get your ticket.